Practice Problems

1.

- Transfer of bus control in either direction takes 250 ns.
- An I/O device with a data transfer rate of 100 KB/sec employs DMA.

a. In burst mode, how long does it take to transfer a block of 512 bytes? (5000500 ns)
b. If data is transferred one byte at a time, calculate the time required for transferring 512 bytes in cycle stealing mode. (5256000 ns)

2.

- Transfer of bus control from processor to device takes 200 ns.
- An I/O device with a data transfer rate of 50 KB/sec employs DMA.
- Transfer of bus control from device to processor takes 150 ns.

a. In burst mode, how long does it take to transfer a block of 128 bytes? (2500350 ns)
b. If data is transferred one byte at a time, calculate the time required for transferring 228 bytes in cycle stealing mode. (4532925 ns)

3.

- Transfer of bus control from processor to device takes 100 ns.
- An I/O device with a data transfer rate of 20 KB/sec employs DMA.
- Transfer of bus control from device to processor takes 150 ns.

a. In burst mode, how long does it take to transfer a block of 256 bytes? (12500250 ns)
b. If data is transferred two bytes at a time, calculate the time required for transferring 256 bytes in cycle stealing mode. (12532000 ns)

4.

- Transfer of bus control from processor to device takes 300 ns.
- An I/O device with a data transfer rate of 40 KB/sec employs DMA.
- Transfer of bus control from device to processor takes 200 ns.

a. In burst mode, how long does it take to transfer a block of 512 bytes? (12500500 ns)
b. If data is transferred four bytes at a time, calculate the time required for transferring 512 bytes in cycle stealing mode. (12564000 ns)

5.

- Transfer of bus control from processor to device takes 100 ns.
- Transfer of bus control from device to processor takes 150 ns.

a. If one of the input/output devices employs DMA in burst mode and takes 10000250 ns to transfer 512 bytes of data, what is the data transfer rate of the device in KB/s? (50 KB/s)

b. Suppose, you are transferring 512 bytes of data in both burst mode and cycle stealing mode. For the first half of the bytes, you use burst mode and for the rest half, you use cycle stealing mode. Assume that in cycle stealing mode, data is transferred 2 bytes at a time. How long will it take to transfer a block of 512 bytes? (Use the data transfer rate found from (a)) (10032250 ns)